Comparative Evaluation of Immersive Virtual Reality and Document-Based Training for Industrial Gas Leak Response

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***Abstract*—Safety-critical operations always require an immediate and trained response. In natural gas processing plants, day-to-day maintenance work and emergency responses are often documented and kept in the form of text-based Standard Operating Procedures(SOPs). These conventional SOPs lack the contextual learning required to perform critical tasks in case of an actual emergency, and thus are susceptible to result in failure of critical systems and man-made disasters. This paper introduces a Virtual Reality(VR) guided-learning experience designed for newly onboarded supervisors and other workers at a gas plant, where on-time action and structured response to a fault are extremely mandatory. The experience incorporates a range of faults that must be identified by the user using audiovisual(AV) cues and SOPs that the user learns in an interactive and immersive embodied way. To evaluate the benefits associated with the proposed approach, a comparative study was conducted in which users were divided into two groups: while one was trained via VR modules, the other group relied on a mentor-guided text-based learning. The results indicate that participants who underwent immersive VR-based guided training achieved significantly higher knowledge retention, recall accuracy, and improved confidence when compared with participants trained using conventional Document-based method. VR-based framework achieved a 13.78% higher post-training accuracy and 45.77% higher mean learning gain than the conventional approach. Source code is given at : https://github.com/Praso28/ICVARS_D046**

***Index Terms*—Virtual Reality, Spatial Cues, Training, Embodiment, Emergency Response**

I. Introduction

Safety training in critical industries is of highest priority and must be completed before induction of crew members and staff of such industries. However, industrial safety training has relied only on traditional methods such as technical manuals, machine documentation, and verbal lectures, which fail to set up the right emergency environment and near-perfect replication of a life-threatening situation in an actual emergency. The 1984 Bhopal gas tragedy, one of the world's worst industrial disasters, is one such example which resulted in thousands of casualties due to lack of emergency response and improper safety training [1]. Similarly, other critical industries continue to suffer significant losses due to improper handling of serious situations. This motivates us to search for a better training method which shifts the learning paradigm from passive to active [2]. We propose an immersive VR-based guided learning framework for industrial gas leak emergency response and evaluate its effectiveness.

The primary contributions of this work are:

- Development of a framework based on immersive safety training for gas leak emergency response in a natural gas processing plant.
- Introduction and integration of industry-specific safety concepts and protocols including Double Block and Bleed (DBB), Lockout-Tagout (LOTO), and multimodal fault localization using spatial audiovisual cues.
- A comparative evaluation of training on the basis of objective assessments, confidence analysis, and statistical validation under a controlled environment for both immersive (VR-based) and traditional (Document-based) methods.

II. Literature Review

A. VR in safety-critical training

Virtual Reality(VR) has emerged as an innovative and reliable medium for supporting trainings and simulations in accident-prone domains without exposing oneself to hazardous conditions. Using immersive interaction and contextual embodiment, a virtually real environment is recreated, and users are allowed to experience procedural workflows in a controlled manner while adhering to realism and a sense of presence. Stefan et al. [3] conducted a systematic review across many safety-critical domains and concluded that VR training can consistently improve user engagement, procedural recall and decision making compared to conventional learning manual-based method. Their works also support the evaluation of pre-test and post-test to measure the true cognitive gains after the exposure to training.

Furthermore, Arif et al. [4] analyzed the benefits of immersive visualization while comparing different rendering devices and their effect on performance during critical situations. The

overall integration of spatial cues, environmental feedback, interactive objects, and real-time decision-making allows oneself to move beyond passive consumption of the material and instead develop an embodied experience and procedural understanding [5].

B. VR in high-risk industries

The application of immersive technologies in high-risk industries such as oil and gas, chemical manufacturing, mining, and emergency responses has gained considerable traction over the last few years. Al-Khafaji et al. [6] explored the role of AR and VR technologies in industrial safety training. The results showed improved hazard recognition, reduced procedural errors, and most importantly, faster reaction times among participants who were exposed to immersive learning stressors.

Mehta et al. [7] further laid the groundwork for understanding the importance of stress factors and responsiveness during procedural immersive training. Traditional document-based learning often fails to reproduce real environmental stressors such as alarms, sensory overload, and visibility limitations. Whereas scenarios involving hazardous gases, time-sensitive operations and confined spaces can be done on a VR system under controlled simulation conditions.

C. Industrial VR frameworks

Some of the recent VR frameworks have focused on recreating a training system integrated with guided interaction, multimodal sensory cues and contextual technical communication. Vercelli et al. [8] proposed an immersive framework, SAMPAI, for industrial procedural training which demonstrated that guided VR training can improve the learner's confidence and increase the retention capabilities of the participant. Similarly, Dubois and Banda [9] reported the successful deployment of VR-based training for hazardous industrial operations in the petroleum industry, and Sonawane et al. [10] demonstrated the beneficial use of immersive technology for explosive neutralization during high-risk scenarios.

Existing VR-based safety training systems have focused primarily on evacuation simulations, hazard awareness, or generic industrial procedures. In contrast, our proposed framework specifically addresses gas leak emergency response in a natural gas processing plant and integrates industry-specific operational safety procedures such as Double Block and Bleed (DBB), Lockout-Tagout (LOTO), and fault localization using spatial audiovisual cues. Furthermore, the proposed framework is supported by a comparative evaluation against conventional document-based training using knowledge assessments, confidence measurements, and statistical data. These characteristics holistically distinguish our proposed framework against the existing literature and address a relatively underexplored intersection of immersive learning and industrial gas leak safety procedures.

III. Problem Formulation

In oil and gas plants, gas processing takes place under high-pressure and safety-critical conditions where even a minor failure may lead to undesirable and dire consequences. Rapid hazard identification, protection equipments, procedural isolation and a clear evacuation plan are key to safe-handling of a emergency situation. This requires careful and intensive training for the workers and supervisors in the plant.

Conventional training employs a rigorous methodical learning which primarily relies on Standard Operating Procedure(SOP) manuals, classroom lectures, and mentor-guided documentation. Although, these instruments cover the theoretical knowledge one must possess, they fail to replicate the urgency, spatial awareness, and contextual learning required during real emergency response executions. This work explores whether immersive VR-based guided learning can improve procedural retention, hazard identification, and learner's confidence compared with traditional document-based training.

A. Problem Statement

The lack of realistic contextual exposure during conventional training can lead to delayed reaction times, incorrect procedural recall, and poor confidence while performing emergency response. In natural gas processing plants, responder must follow the SOPs and react in a time-bound manner to crucial tasks such as Self-Contained Breathing Apparatus(SCBA) deployment, leak isolation, Double Block and Bleed(DBB) Protocol, and Lockout-Tagout(LOTO) implementation.

The primary challenge addressed in this work is to determine whether VR-based training can provide measurable and beneficial improvements over conventional text-based instructions.

B. Hypothesis

In this study, the hypotheses are defined as follows :

- H_1**:** VR-based training results in higher procedural recall, hazard identification and confidence than Document-based training.
- H_0**:** There is no statistically significant difference between VR-based training and Document-based training.

IV. Experimental Design

A. Objective

The primary objective of this study is to evaluate the effectiveness of VR-based training for industrial gas leak scenario. Specifically, this work compares immersive VR-based learning with mentor-guided document-based learning in terms of procedural recall, hazard identification, fault isolation, compliance of SOPs and confidence improvement.

B. Participants

A total of 30 individuals voluntarily participated in the study. All the individuals were between 18 to 24 years of age and had no prior industrial safety operations experience. The participants were further randomly divided into two groups:

- **Group A:** Document-Based Training $(n = 15)$
- **Group B:** VR-Based Training $(n = 15)$

Anonymized records were maintained and all participants provided informed consent prior to participation.



Fig. 1. Experimental setup and representative scenes from the proposed immersive VR-based training framework. Clockwise from top: in-environment procedural guidance and emergency SOPs, participant interaction during controlled evaluation, execution of the Double Block and Bleed (DBB) isolation procedure for fault mitigation, and the simulated industrial Acid Gas Removal Unit (AGRU) facility used during training.

C. Hardware + Software

The VR-training model was developed with Unreal Engine 5.4.4 on a desktop computer equipped with a 2 TB Solid State Drive (WD Green SN350), GeForce RTX A5000 NVIDIA GPU, and a Intel Core i9-13900K processor. The simulation was then deployed and tested on Meta Quest 3 head-mounted display which offered a high resolution of 2064 $\times$ 2208 pixels per eye and a 110° field of view, providing subjects with a clear and immersive visual experience.

V. VR Training Environment

The proposed training environment extends an industrial Acid Gas Removal Unit (AGRU) as its base inside a natural gas processing facility. The overall flow is designed while keeping in mind an immersive learning experience for the user where it is possible to interact with the equipments and develop a sense of presence at the emergency site under controlled conditions. A systematic illustration is shown in Fig. 2, which highlights the flow of events in the VR-based learning model.

A. Supervision Room

The learner starts in the hallway and navigates to the supervision room after the full plant tour cutscene is completed. This is the first room a learner is introduced to and the place where contextual instructions, protocols, information and emergency SOPs exist.

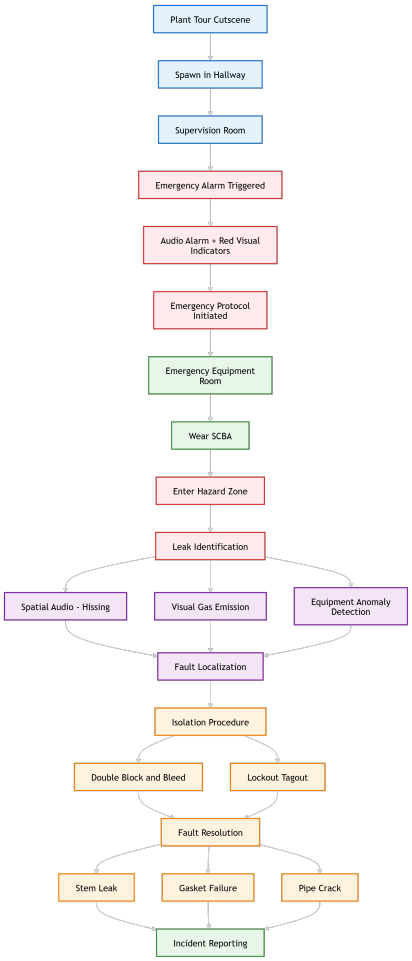


Fig. 2. Workflow of the proposed immersive VR-based industrial gas leak training environment. Participants sequentially perform contextual orientation, emergency equipment deployment, leak localization, fault isolation, and incident reporting using guided multimodal interaction.

B. Alarm Event

After the learner is done familiarizing with the environment, an emergency event is triggered. A gas leak detected warning appears, and synchronized auditory alarms and red visual warning goes off simultaneously. Immediate instructions are given and learner is asked to follow the SOP.

C. Emergency Equipment

Learner is instructed to prioritize safety at all times and correctly wear a Self-Contained Breathing Apparatus (SCBA) before entering any kind of hazardous operational zone.

D. Leak Identification

Learner navigates to the leak using multi-modal sensory cues including spatial audio, hissing sound, and visual gas emission effects. Identification of the fault location must be done before moving to the isolation procedure.

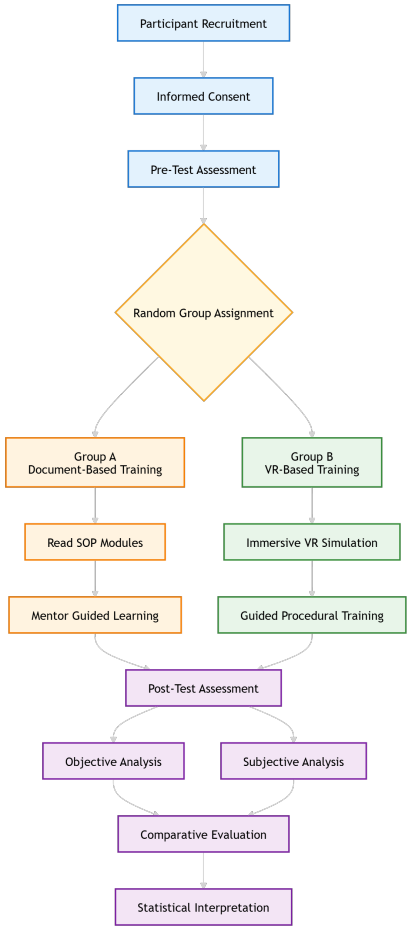


Fig. 3. Experimental protocol comparing mentor-guided document-based learning and immersive VR-based training. Participants complete pre-test assessment, training intervention, post-test evaluation, and confidence analysis.

E. Isolation Procedure

After the learner has successfully found the fault location, the isolation procedures must be completed step-by-step. It involves initiating Double Block and Bleed (DBB) Protocol, Lockout-Tagout (LOTO), fault identification, and incident reporting. There are 3 fault scenarios covered in the simulation, namely - stem leakage, gasket failure, and pipe cracking.

VI. Experimental Protocol

The overall experiment was conducted in 3 sequential phases: pre-test assessment, training modules (either VR-based or Document-based, as per group assignment), post-test assessment. At first, the participant underwent pre-test assessment, which was designed to evaluate the baseline knowledge of the participant related to the industrial safety operations. It consisted of 15 objective-type questions covering key industrial concepts such as Self-Contained Breathing Apparatus (SCBA) deployment, fault localization, hazard identification, procedural SOPs, DBB protocol execution, LOTO application and emergency decision-making. It also comprised of 3 subjective questions covering confidence levels in different kind of emergency situations relating to our study.

Following the assessment, the participants were randomly assigned to a group. Group A was trained with mentor-guided document-based learning, whereas Group B completed their immersive training with VR-based modules. After the training was concluded, participants faced an isomorphic post-test assessment containing conceptually equivalent but rephrased questions to minimize recall bias while maintaining a comparable level of difficulty. Both groups were subjected to a similar controlled environment, and the training system was designed to provide a comparable exposure to the learning material and operational procedures. While Group A was assigned a mentor who guided through documentation, Group B participated in an immersive simulated training using a VR headset.

VII. Evaluation Metrics

The effectiveness of the proposed training framework was evaluated using both objective and subjective performance measures.

A. Knowledge Accuracy

Participant test performance was normalized with respect to the maximum attainable score of 15: $K = \frac{S}{15} \times 100$, where S denotes the participant score obtained during assessment.

B. Learning Gain

Improvement in procedural understanding was quantified as the difference between post-test and pre-test scores: $G_k = S_{post} - S_{pre}$, where S_{pre} and S_{post} represent the participant's scores before and after training respectively.

C. Confidence Improvement

Subjective confidence improvement was evaluated using average Likert-scale ratings collected before and after training: $G_c = C_{post} - C_{pre}$, where C_{pre} and C_{post} denote the average participant confidence ratings on a five-point scale.

D. Uncertainty Measure

Initial uncertainty was quantified using the frequency of "I Don't Know" responses during pre-test assessment: $U = \frac{DK}{15} \times 100$, where DK denotes the number of uncertain responses.

VIII. Subjective Test Analysis

The subjective effectiveness of the proposed training system was evaluated on the basis of collected confidence scores on the five-point Likert scale in the pre-test and post-test assessments. *As illustrated in Fig. 4*

Participants in Group A demonstrated an increase in average confidence score from 1.71 before the document-based training to 3.13 after the training. Similarly, Group B demonstrated an even better and more robust increase in average confidence score from 1.91 prior to immersive training to 3.98 after completion of the training. This corresponds to a significant gain of 2.07 points on a five-point Likert scale. The results provide insight to how embodied experience and guided learning can

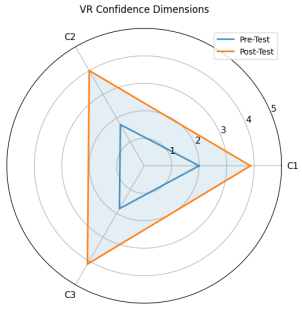


Fig. 4. Question-level confidence profile of the VR-based training group across the three subjective evaluation dimensions. A consistent increase in confidence was observed following immersive procedural training.



Fig. 5. Comparison of average knowledge accuracy and confidence levels before and after training across both experimental groups. The immersive VR-based group demonstrated higher post-training performance across both objective and subjective metrics.

lead to better confidence scores and a sense of achievement in the participant.

IX. Objective Test Analysis

The objective performance was measured using the scores of pre-test and post-test assessments. *As illustrated in Fig. 5* For the Document-based training group, the average pre-test assessment score turned out to be 3.53 out of 15 which improved in the post-test assessment by 6.87, leading to a final average post-test assessment score of 10.40. On the other hand, the VR-based training group scored an average of 3.13 out of 15, which signifies the limited understanding of safety operations before the training. Following the immersive training, the average post-test assessment score increased fourfold to 12.47, corresponding to a mean gain of 9.34 points.
Additionally, during the pre-test assessment, participants selected the "I Don't Know" option 43.56% of the times which reflected on the uncertainty observed when participants were evaluated without prior training in safety-critical operations. *This uncertainty is illustrated in Fig. 6*

X. Comparative Analysis

A comparative evaluation was done between the Group A and Group B results to understand the performance and assess the differences observed in terms of procedural understanding, confidence, recall accuracy and knowledge gain. *As illustrated in Fig. 7* and a summarized comparison of group-wise learning performance, confidence improvement, and statistical significance is presented in Table I.

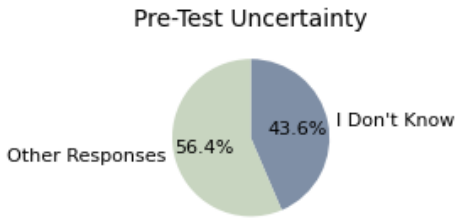


Fig. 6. Pre-test response distribution across all participants.

Metric	Document-Based	VR-Based
Participants (n)	15	15
Pre-Test Score	3.53	3.13
Post-Test Score	10.40	12.47
Knowledge Gain	6.87	9.34
Post Accuracy (%)	69.33	83.11
Confidence Gain	1.42	2.07
Welch's t-test (p)	$p < 0.01$	

TABLE I
SUMMARY OF GROUP-WISE LEARNING, PERFORMANCE AND CONFIDENCE IMPROVEMENT

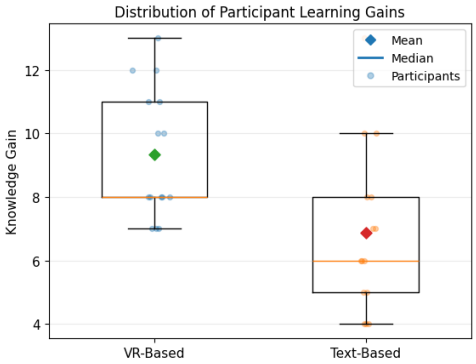


Fig. 7. Distribution of participant-level learning gains across both training groups. The VR-based group demonstrates higher median improvement and more consistent procedural learning gains.

The comparative analysis between the performance of both groups highlights clear differences in terms of procedural recall, confidence development, and knowledge gains. Participants who underwent Document-based training showed an average knowledge gain of 6.87 points, leading to a post-training accuracy score of 69.33%. In contrast, Group B participants who underwent VR-based training achieved a higher average knowledge gain of 9.34 points and a post-training accuracy of 83.11%. The average confidence gain also followed a similar trend where the confidence gain for the Document-based group was reported at 1.42 points and VR-based group stood at a higher mark of 2.07 points on the five-point Likert Scale.
Further, an independent two-sample Welch's t-test on the learning gains supported our hypothesis and the VR-based

group was found to be statistically different from the Document-based group ($t = 2.89$, $p < 0.01$). The observed effect size was large ($d = 1.05$), suggesting a substantial practical advantage of immersive VR training over traditional methods. The mean difference in learning gain between the two groups was 2.47 points, with a 95% confidence interval of [0.71,4.22]. The confidence interval indicates that the observed improvement is unlikely to be due to sampling variability as it excludes zero and consistently favors the VR-based training approach. The findings of our study collectively support the hypothesis that immersive VR-based training can improve procedural understanding, demonstrate stronger confidence development, and better knowledge retention compared with conventional document-based learning. Thus, null hypothesis (H_0) was rejected and the alternative hypothesis (H_1) was supported.

XI. Discussion and Limitations

The results produced in our study indicate that immersive VR-based training for safety-critical industries can substantially improve the hazard recognition capabilities, deep procedural understanding of the underlying situation and improve overall confidence during emergency responses. These indicated improvements can be linked to the usage of embodied experience, audio-visual spatial cues, contextual learning and active engagement of an individual during the training period [11]. Unlike passive methods like Document-based learning, VR-based training allowed trainees to experience the critical near-realistic emergency situation safely. An important aspect highlighted by this study is the practical utility of the proposed VR-based guided training framework. The model demonstrates strong applicability for industrial safety training programs. It offers a scalable and repeatable training solution that can reduce dependency on physical mock drills, minimize operational disruptions, and lower training costs over time.

Despite the strong results, the limitations of this study included participants who were mostly students belonging to 18-24 age group with no prior safety operations experience or knowledge. Trained operators, supervisors, or field technicians working in real gas-processing facilities remained unaccounted for in the study. This limits the immediate generalizability of our proposed framework and the findings can only be interpreted within the context of an academic population. Future studies should include larger and more diverse participant groups across different age groups, professional backgrounds, and experience levels. Another limitation of the study is that only immediate post-training outcomes were evaluated. The study does not account for the long-term retention of procedural knowledge, confidence, and operational efficiency following either training intervention. Future work should incorporate an extended framework to evaluate the long-term effectiveness of VR-based training.

XII. Conclusion

This work presents an immersive VR-based guided training framework for industrial gas leak emergency response and demonstrates its effectiveness against conventional document-based passive methods of training like manuals, class-lectures and written SOPs. The findings confirm that VR-based training enhances procedural understanding, improves hazard recognition, increases learner confidence, and contributes to better safety preparedness. Experimental results point towards the benefits observed in terms of procedural understanding, confidence gain in learner, and safety operational readiness following immersive training. Future work will include real-time haptic feedback integration [12], multi-user collaborative environments [13] to simulate team-based emergency response, and deployment in real industrial training environments.

XIII. Acknowledgment

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